Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.5 Volume

Volume

STARTER · 5 MIN

Date: _____ **Title: Volume**

LEARNING OBJECTIVES

- Calculate the volume of a cube or a cuboid, and convert between cm^3, ml and litres

STARTER *Answer in your book. No calculators.*

1. Work out $2 \times 2 \times 2$
2. Work out 3^3
3. Work out $6 \times 10 \times 4$
4. How many ml are in 1 litre?

Starter — Answers

STARTER

Tick or correct in green pen

1. **8**
2. **27**
3. **240**
4. **1000 ml**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use the volume formulas for a cube and a cuboid	☐
SILVER	I can convert between cm^3, ml and litres	☐
GOLD	I can find missing sides and solve problems in real contexts	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Volume

INTRODUCTION · 3 MIN

Volume is the amount of space inside a 3D solid.

Cuboid

length × width × height

$V = lwh$

Cube

side × side × side

$V = l^3$

Units

Volume is in cubic units.

cm^3, m^3

Capacity

1 ml = 1 cm^3

1 litre = 1000 cm^3

Examples 1 and 2 — Cubes and Cuboids

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

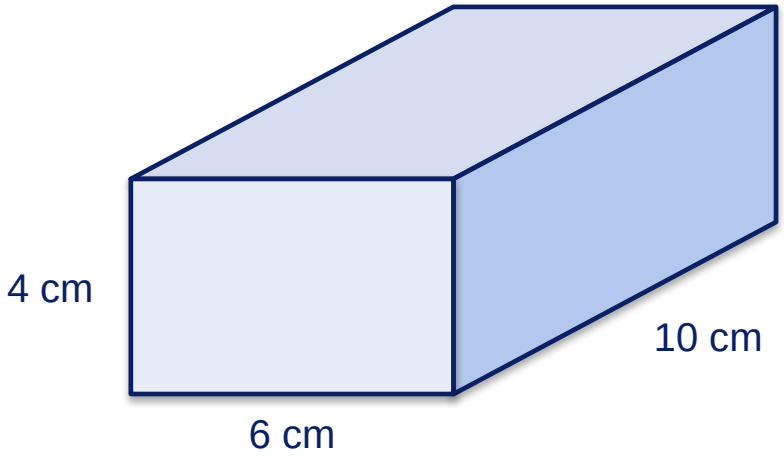
Hint Cuboid: $V = lwh$ Cube: side^3

Example 1

Find the volume of a cube of side 3 cm.

Example 2

Find the volume.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

side × side × side

$V = 3 \times 3 \times 3$

$= \mathbf{27\ cm^3}$

Tip: $3^3 = 27$, not 9.

Example 2

length × width × height

$V = 6 \times 10 \times 4$

$= 60 \times 4$

$= \mathbf{240\ cm^3}$

Tip: the units are cm^3.

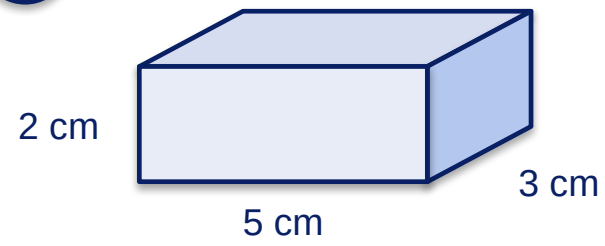
Task 1

PRACTICE · 9 MIN

BRONZE *Find the volume*

Hint Cuboid: $V = lwh$ Cube: side^3

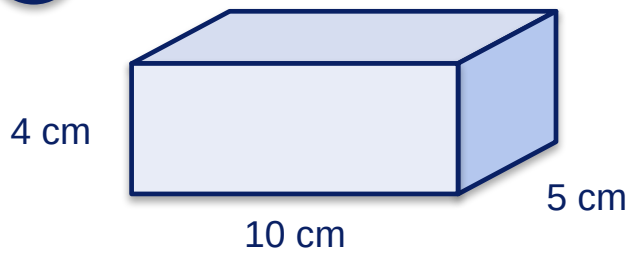
1 Volume



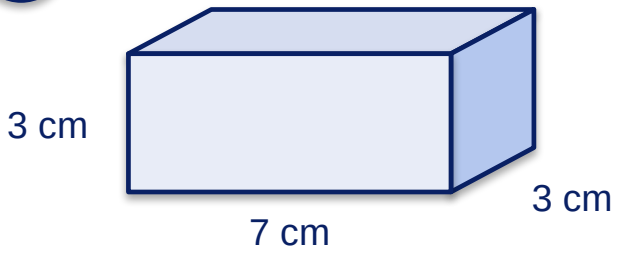
2 Volume



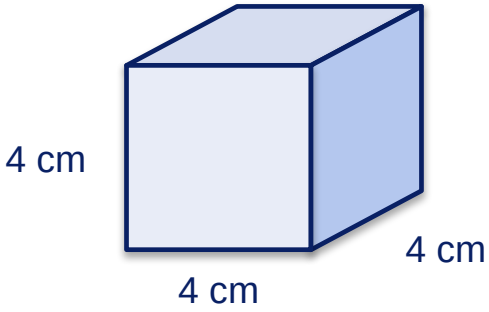
3 Volume



4 Volume



5 Volume



★ Volume



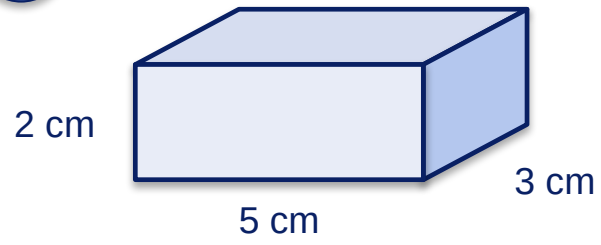
Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

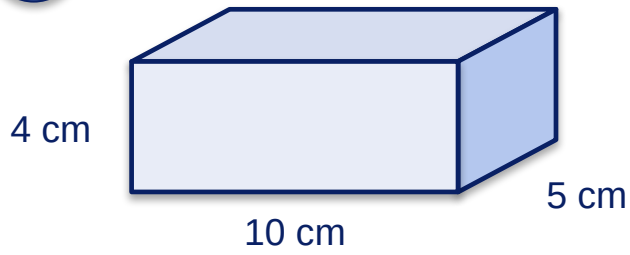
1 **30 cm³**



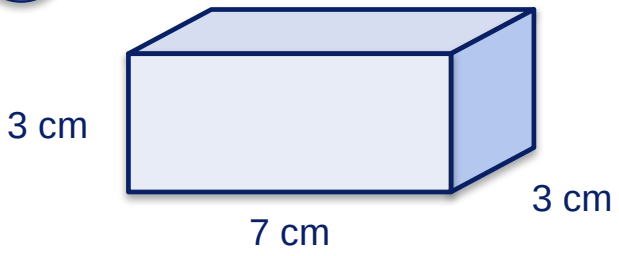
2 **48 cm³**



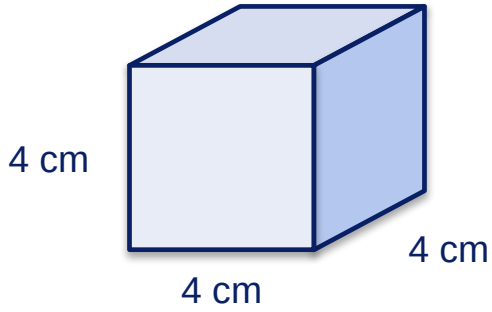
3 **200 cm³**



4 **63 cm³**



5 **64 cm³**



★ **150 cm³**



Examples 3 and 4 — Volume and Capacity

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint 1 ml = 1 cm^3 1 litre = 1000 cm^3

Example 3

A box is 3 cm by 4 cm by 3 cm. Find its volume in cm^3 and in ml.

Example 4

Convert 1.2 litres to cm^3.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

1 cm³ holds 1 ml

$V = 3 \times 4 \times 3$

$= 36 \text{ cm}^3$

$= 36$ ml

Tip: cm³ and ml are the same size.

Example 4

1 litre = 1000 cm³

1.2 litres $= 1.2 \times 1000$

$= 1200 \text{ cm}^3$

Tip: litres to cm³: × 1000.

Task 2

PRACTICE · 9 MIN

SILVER *Convert*

Hint 1 ml = 1 cm^3 1 litre = 1000 cm^3

1. 250 cm^3 to ml
2. 5000 cm^3 to litres
3. 1.5 litres to cm^3
4. 750 ml to litres
5. 6000 cm^3 to litres

★ Is 3500 cm^3 more than 3.2 litres?

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **250 ml**
2. **5 litres**
3. **1500 cm³**
4. **0.75 litres**
5. **6 litres**

★ **Yes: 3.2 litres = 3200 cm³**

Examples 5 and 6 — Missing Lengths and Capacity

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

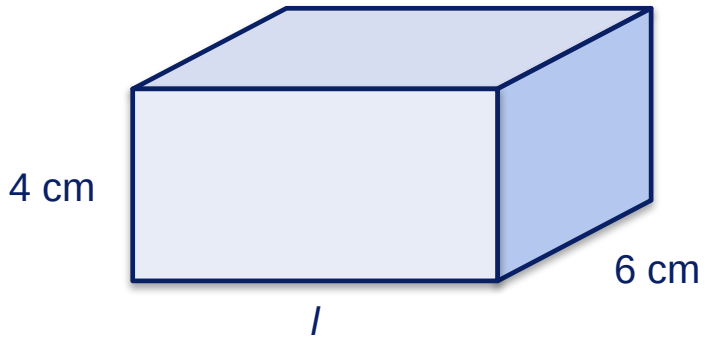
Hint Write an equation Litres: ÷ 1000

Example 5

(2 marks)

The volume is 168 cm³.

Find the length *l*.



Example 6

(3 marks)

A tank is 30 cm by 20 cm by 10 cm.

How many litres does it hold?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Write an equation

$l \times 6 \times 4 = 168$

$24l = 168$

$\mathbf{l = 7\ cm}$

Tip: 168 ÷ 24 = 7.

Example 6

Volume, then litres

$V = 30 \times 20 \times 10$

$= 6000\ cm^3$

$\mathbf{= 6\ litres}$

Tip: divide by 1000 for litres.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: volume problems

Hint Write an equation Litres: ÷ 1000

1. Volume 120 cm³, 5 cm by 4 cm by *h* cm. Find *h*.

2. Cube volume 64 cm³. Side?

3. Cube volume 125 cm³. Side?

4. Tank 50 cm by 40 cm by 30 cm. How many litres?

5. Box 10 cm by 10 cm by 10 cm. How many litres?

★ Carton holds 0.5 litres. Base 10 cm by 5 cm. Height?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $h = 6$ cm
2. 4 cm
3. 5 cm
4. 60 000 cm^3 = 60 litres
5. 1 litre

★ 500 ÷ 50 = 10 cm

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Volume of a cuboid 3 cm by 4 cm by 5 cm

SILVER Convert 2.5 litres to cm^3

GOLD Convert 750 cm^3 to ml

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **60 cm^3**

SILVER **2500 cm^3**

GOLD **750 ml**

Exam Practice — Volume

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

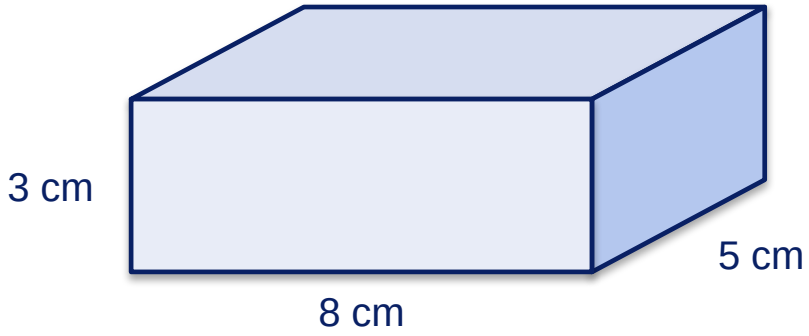
Hint Cuboid: $V = lwh$ Cube: side^3

Question 1

(2 marks)

Work out the volume of this cuboid.

Diagram NOT accurately drawn



..... cm^3

Question 2

(2 marks)

A cube has edges of 6 cm.
Work out its volume.

..... cm^3

Exam Practice — Capacity

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint 1 ml = 1 cm^3 1 litre = 1000 cm^3

Question 3

(2 marks)

Convert

(a) 4500 cm^3 to litres (1)

..... litres

(b) 0.3 litres to ml (1)

..... ml

Question 4

(3 marks)

A fish tank is 40 cm by 25 cm by 30 cm.

How many litres does it hold when full?

..... litres

Exam Practice — Volume Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Write an equation Litres: ÷ 1000

Question 5

(3 marks)

A cuboid has volume 360 cm³.
It is 12 cm long and 5 cm wide.

Work out its height.

..... cm

Question 6

(4 marks)

A carton is 6 cm by 6 cm by 20 cm.
It is full of juice.

How many full 150 ml glasses can be poured from it?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$V = 8 \times 5 \times 3$ **M1**

$= 120\ \text{cm}^3$ **A1**

Tip: multiply all three lengths.

Question 2

$V = 6 \times 6 \times 6$ **M1**

$= 216\ \text{cm}^3$ **A1**

Tip: 36 × 6 = 216.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 4.5 litres B1

(b) 300 ml B1

Tip: 1 litre = 1000 ml = 1000 cm³.

Question 4

$V = 40 \times 25 \times 30$ M1

$= 30\,000\ \text{cm}^3$ A1

$= 30$ litres A1

Tip: divide by 1000.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$12 \times 5 \times h$	$= 360$	M1
$60h$	$= 360$	M1
$\boldsymbol{h}$	$\mathbf{= 6\ cm}$	A1

Tip: 360 ÷ 60 = 6.

Question 6

V	$= 6 \times 6 \times 20$	M1
	$= 720\ cm^3 = 720\ ml$	A1
glasses	$= 720 \div 150 = 4.8$	M1
	$\mathbf{= 4\ glasses}$	A1

Tip: only 4 glasses are full.